

EXPERIENCE

The Home Depot, Machine Learning Engineer II

Oct 2024 - Current

- Designed and implemented a scalable system architecture for a Databricks proof of concept, providing seamless version control for data scientists and merchants through an automated CI/CD pipeline that directly influenced strategic decisions by the EVP/CIO.
- Spearheaded the creation of the "NL2SQL" functionality of an internal Gen BI Chatbot, allowing data partners to gain insights on real data using natural language.
- Developed and productionized foundational Dataform pipelines for two strategic ML partnerships, enabling order optimization and segmenting 54 million Home Depot customers into 5 granular sub-segments for real-time, data-driven personalization.
- Engineered and maintained an automated weekly updates process, slashing manual effort by over 97% (from 6 hours to 10 minutes per week) and saving approximately 303 company hours annually.
- Championed team-wide adoption and enablement of Dataform by delivering regular knowledge transfers and creating scalable resources, increasing team self-sufficiency and reducing recurring support.

The Home Depot, Software Engineer I

July 2023 - Oct 2024

- Led the metadata auditing workstream, scaling a proof of concept into a production-ready solution, now impacting multiple teams enterprise-wide.
- Enabled Dataform adoption across 150+ GCP projects, providing technical support and creating a template repository with built-in CI/CD, speeding up productionization for users.
- Created AI enabled automation scripts that reduced manual task times by up to 90%, cutting a repeatable process from 1 hour to 5 minutes.
- Contributed shared GAR components that get used in an ML Anomaly Detection process across all EDW data daily.
- Organized and hosted team-wide meetings, including greater team celebrations and staff meetings.

Computer Science Instructor, Nova Academy

Oct 2021 - Feb 2022

Nova Academy is an education network.

- Curated and taught a curriculum covering basic programming topics and fundamental engineering skills to ignite STEM interest in young developers. Topics covered include conditionals, variables, and functions.

PROJECTS

OpenTuner

Feb 2026 - Current

OpenTuner is an IOS Guitar Tuner and Metronome.

- Spearheaded the end-to-end development of "OpenTuner," an iOS application, achieving a 5-star rating on the App Store for its user-centric and ad-free experience.
- Engineered a custom real-time pitch detection algorithm using Swift and Core Audio, optimizing it for a diverse range of acoustic and electric instruments.
- Drove the full product lifecycle, from initial concept and UI/UX design in Figma to deployment and iterative updates based on user feedback.
- Enhanced the app's feature set by implementing a professional-grade metronome and a system for users to create and save custom instrument tunings.

EDUCATION

BS, Computer Science, Loyola Marymount University - Los Angeles, CA (2022)

- Cum Laude, LMU Arrupe Scholarship (2018-22), Dean's List, Phi Delta Theta

MS, Computer Science, Loyola Marymount University - Los Angeles, CA (2023)